

NEPAL COLLEGE OF INFORMATION TECHNOLOGY
Mid-Term-Spring-2026

Level: Master

Programme: ME/CS

Course: Artificial Intelligence

Year: 2026

Full Marks :60

Time : 2 hrs

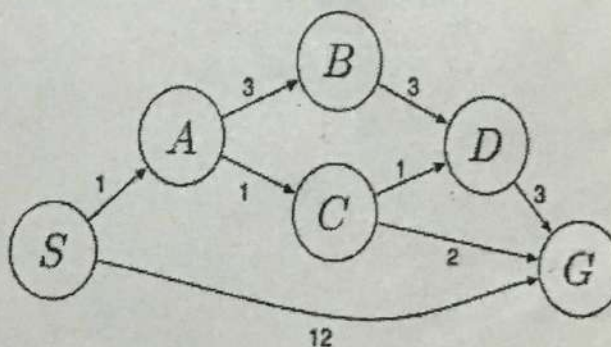
Candidates are required to give answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions

- 1 a) What is Artificial Intelligence? If the Turing Test is passed, does this show that computers exhibit intelligence? State your reasons. 8+7
- b) For following graph, compute A* search. Assume start state is S and goal state is G.

State	h_1
S	5
A	3
B	6
C	2
D	3
G	0



- 2 a) You are given two jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug? Solve by production rule system. 7+8
- b) Write an algorithm for Backtracking Search for CSPs. Use backtracking to solve following CSP

BASE
 + BALL

 GAMES

3 a) Compare uninformed and informed search technique in terms of space and time complexity. 8+7

b) Explain Deep learning and what are their applications.

4 a) Suppose you are hired by ecommerce company of Nepal to develop a chat bot to answer the queries posted by its customers through its website. 7+8

i. What do you think will be the issues with text complains posted.

ii. Discuss about the steps of NLP to solve this problem.

iii. Can we use LLMs in this scenario?

b) Write short notes on:

i Learning

ii Knowledge and it's Representation